

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch: Electronics and Tele./Electronics & Communication

Semester : VII

Subject Code & Name: BTETC701 & Microwave Engineering

Max Marks: 60

Date:02-01-2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level /CO)	Mark s
Q. 1 Solve Any Two of the following.	CO1	12
A) Poynting theorem states that the total complex power fed into a volume is equal to the algebraic sum of the active power dissipated as heat, plus the reactive power proportional to the difference between time average magnetic and electric energies stored in the volume, plus the complex power transmitted across the surface enclosed by the volume. Explain and Prove this statement		6
B) Prove that Skin Depth is $\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\alpha} = \frac{1}{\beta}$		6
C) Certain transmission line has a characteristic impedance of $75 + j0.01 \Omega$ and is terminated in a load impedance of $70 + j50 \Omega$. Compute (a) The reflection coefficient; (b) The transmission coefficient. (c) Verify the relationship $T^2 = \frac{Z_L}{Z_0} (1 - \Gamma_L^2)$ (d) Prove that the transmission coefficient equals the algebraic sum of 1 plus the reflection coefficient		6
Q.2 Solve Any Two of the following.	CO1, CO2	12
A) An air-filled rectangular waveguide of inside dimensions 7×3.5 cm operates in the dominant TE ₁₀ mode a. Find the cutoff frequency. b. Determine the phase velocity of the wave in the guide at a frequency of 3.5 GHz. c. Determine the guided wavelength at the same frequency.		6
B) Explain the Two hole Directional Coupler		6
C) Design the four post circulator using Magic Tee		6
Q. 3 Solve Any Two of the following.		12
A) Explain Construction and Principle of operation of 8 cavity cylindrical travelling wave magnetron	CO3,	6

B)	Explain the process of velocity modulation and bunching process, Applegate diagram of Cylindrical Magnetron	CO4,	6
C)	Explain the process of avoiding the Mode jumping of Cylindrical Magnetron	CO5	6
Q.4	Solve Any Two of the following.	CO6,	12
A)	Explain the S - parameter measurement of the Microwave	CO7	6
B)	Explain the frequency measurement		6
C)	Explain the Low, Medium and High Power measurement		6
Q. 5	Solve Any Two of the following.	CO6	12
A)	Explain Microstrip line in details		6
B)	Explain the Parallel Strip line and Coplanar Strip line		6
C)	What are Hazards of Electromagnetic Radiation		6

***** End *****